



CARPENTRY

HPC Carpentry

An Overview of HPC Carpentry's Activities

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A Very Brief History of The Carpentries

- ▶ Software Carpentry was founded in 1998 with the mission of teaching lab skills for research computing by Greg Wilson and Brent Gorda.
- ▶ HPC Carpentry making an appearance in 2013
- ▶ Since then several presentations and workshops
- ▶ Formally accepted to Lesson Program Incubation, June 2023



Dr Greg Wilson
<https://carleton.ca/scs/?p=14196>

- ▶ Core curriculum: Using High-Performance Computing Systems, HPC Workflow Management with Snakemake
- ▶ Recommend to start with the SWC Unix Shell lesson where learners are not already familiar with working in the shell
- ▶ An extended curriculum of other lessons



Current Status: Lesson Support

- ▶ Instructor Onboarding documentation available
- ▶ Lesson Maintainers onboarded by The Carpentries
- ▶ Infrastructure support allows customisation of lessons to reflect local cluster configurations



Current Status: Community

- ▶ Welcomed many new members to regular community calls + Maintainer teams
- ▶ Gathered interest in teaching+hosting workshops from the wider Carpentries community
- ▶ Completed Steering Committee elections



Current Status: Lesson Program Adoption

- ▶ Adoption as an official lesson program of The Carpentries was planned for 1 September 2026
- ▶ Changes in Carpentries Core Team support called for a short delay
- ▶ Adoption will be completed between now and November 2026

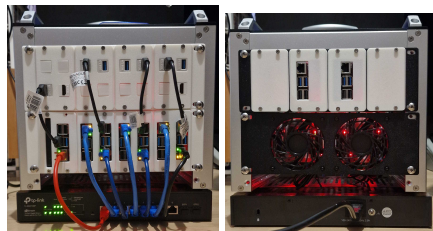
Running Workshops

- ▶ Workshops since January 2025
 - ▶ NIST – 2
 - ▶ Newcastle – 5
- ▶ Feedback to improve lesson infrastructure for adaptation to a specific HPC
- ▶ Snippets
- ▶ Episode on array jobs now included
- ▶ Onboarding for instructors



A miniHPC - CarpentriesOffline

- ▶ Pixie: Raspberry Pi 4s, one login node, five compute nodes, one access point
- ▶ The same open source software stack as many HPCs: Slurm, lmod, munge
- ▶ Running Workshops and Hackerthons
 - ▶ Lesson to build a login node and a compute node
 - ▶ Adapt HPC lesson for Pixie
 - ▶ Networking lesson



Front and Back

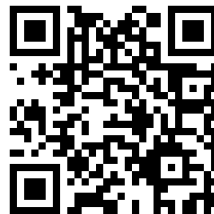


Comms Channels

- ▶ Come talk to me
- ▶ Visit our website
- ▶ Join our bi-weekly meetings
- ▶ Join our channel in The Carpentries' Slack workspace



HPC Carpentry



CarpentriesOffline



CARPENTRY

Your Help is Needed

- ▶ Do you have any suggestions for more domain-specific conferences where we could reach members of HPC Carpentry's target audience of learners and or potential instructors hosts for workshops?
- ▶ Are you interested in teaching, organising and/or hosting HPC Carpentry workshops?
- ▶ Are you interested in getting involved with maintaining the lessons?



- ▶ **Greg Wilson's talk at PyCon 2014**
<https://www.youtube.com/watch?v=Wv0rf3Fyh7M>
- ▶ **Andrew Reid's talk at SIGHPC 2024**
<https://www.youtube.com/watch?v=FtKO619O5g0>
- ▶ **Peter Steinbach's blog post, HPC in a day?**
<https://carpentries.org/blog/2017/06/hpccarpentry>
- ▶ **HPC Carpentry Website** <https://www.hpc-carpentry.org>
- ▶ **CarpentriesOffline for the miniHPC** <https://carpentriesoffline.org>
- ▶ ChatGPT for its help with the $\text{\LaTeX}$ template!